

Petras Vestartas

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PERSONAL

Neufeldstrasse 19

3012 Bern, Switzerland

Born 13/12/1989, Lithuanian

LANGUAGES

English **C1**

German **B1**

French **A1**



// COMPETENCES

Open-Source Software

C++, Python, Rust and C# libraries with GitHub CI for docs, tests and builds on Windows, macOS, Linux.

Timber Construction

Design-to-fabrication workflows for highly complex geometry in subtractive manufacturing.

Robotics

Commissioning and kinematics for ABB 6700 on a 6.7 m track, 5-axis CNCs, lidar, cameras, grippers.

Publications & Funding

Conference and journal papers, grant writing and coordination with multidisciplinary research groups.

Teaching

EPFL and ETHZ courses, lectures, COMPAS workshops at AA Seoul and Zaha Hadid Architects.

Architectural Design

Trained architect with strong representation and planning skills, international competitions.

// EXPERIENCE

2023–NOW

Post-doc, Block Research Group (BRG), ETH Zurich, Switzerland.

Development of the COMPAS framework, with C++ and Fortran90 bindings for Python, research in timber vaulted floors, workshops, developer and industry days, brg.ethz.ch

2021–2023

Post-doc, Laboratory for Timber Constructions (IBOIS), EPFL, Lausanne, Switzerland.

Development of the Cockroach point-cloud processing package and the `compas_wood` package; commissioned the ABB 6700 robot and 6.7 m track (CHF 227k equipment funding), supervised a PhD thesis, epfl.ch/labs/ibois_dfab.ch

2016–2017

Research Assistant, CITA, Royal Danish Academy of Fine Arts (KADK), Copenhagen, Denmark.

Flora Robotica, Complex Modelling 5 Inflated Restraint project, royaldanishacademy.com/CITA

2014–2016

Architect, DMAA, Vienna, Austria. Contributed to winning of competitions and commissions in Austria, Italy, Germany and China, dmaa.at

2012–2013

Architect, Do-Architects, Vilnius, Lithuania. Vilnius Bajorai masterplan, doarchitects.lt

2011

Intern, CEBRA, Aarhus, Denmark. Urban planning, cebraarchitecture.dk

2010

Intern, Vakarinis Fasadas, Siauliai, Lithuania. Contributed to commissioned projects.

2009

Furniture Designer, Graniteka, Siauliai, Lithuania. 3D preparation of stone furniture for fabrication

// EDUCATION

2017–2021

Ph.D., Laboratory for Timber Constructions (IBOIS), École Polytechnique Fédérale de Lausanne (EPFL), Switzerland.

2013–2014

Erasmus Exchange Program, Royal Danish Academy of Fine Arts (KADK), Copenhagen, Denmark

2012–2014

Master of Science (M.Sc.) in Architecture, Vilnius Academy of Arts (VAA), Lithuania

2008–2012

Bachelor of Science (B.Sc.) in Architecture with distinction, Vilnius Academy of Arts (VAA), Lithuania

// SOFTWARE

Own tools

	<u>session</u> C++ Python Rust : one geometry kernel, three languages: same API, same tests <u>wood / compas wood</u> C++ Python : finds where plates touch and cuts fitting joints into both <u>OpenNest</u> C# C++ <u>compas nest</u> C++ Python : 2D nesting for CNC and laser cutting <u>NGon</u> C# : polygonal mesh processing for Grasshopper
IBOIS	<u>Cockroach</u> C++ C# : point clouds with CGAL, Open3D and Cilantro <u>COMPAS ABB6700 IRBT</u> Python ROS : ABB 6700 robot on a linear track <u>Raccoon</u> C# : 5-axis CNC G-code
RHINO	<u>Fox</u> C# <u>Scatter</u> C# <u>Mesh Curvature</u> C# <u>Boundary First Flattening</u> C#

Contributions

COMPAS	<u>compas cgal</u> C++ Python : CGAL bindings on PyPI and conda-forge <u>compas occt</u> C++ Python : OpenCASCADE BRep and NURBS backend <u>compas libigl</u> C++ Python <u>compas shapeop</u> C++ Python <u>compas</u> Python <u>compas nanobind package template</u> C++ Python : template for C++ extensions
BRG	<u>compas model</u> Python : one model for design, analysis and fabrication <u>RhinoVAULT</u> Python <u>compas grid</u> Python <u>compas cra</u> Python <u>compas lmgc90</u> Fortran90 C++ Python : LMGC90 discrete element solver bindings <u>compas manifold</u> C++ Python : guaranteed-manifold mesh booleans

Development Stack

C++	CGAL, OpenCASCADE, libigl, Eigen, Boost, Open3D, PMP, ShapeOp, OpenGL
PYTHON	nanobind and pybind11 bindings, COMPAS, Rhino3D, PyPI and conda-forge
RUST	own geometry kernel, WebGPU (WGPU)
C#	Rhino and Grasshopper plugins, professional and teaching use
OTHER	Fortran90 (LMGC90), JavaScript (Vue.js), Java (Processing), Protobuf, CMake, Git
CAD	Rhino3D / Grasshopper, Revit, AutoCAD, Adobe, ArcGIS, V-Ray

// AWARDS & GRANTS

2026	ETHZ D-ARCH Enabling Grant , to prepare larger funding bids such as Innosuisse	CHF 15'000
2023	ETHZ Partnership Council for Sustainable Construction and Digital Fabrication , Timber Vaulted Floor materials and CNC fabrication	CHF 19'000
2022	ENAC Equipment Call, EPFL , high-precision scanners, robotic integration parts	CHF 33'949
2022	Best Paper Award – Runner-Up , CAADRIA, for the Cockroach point-cloud paper	
2020–2022	EPFL Equipment Funding , ABB 6700 robot and 6.7 m track, specifications with integrators and suppliers, spindle, tool-changer and multi-move documents	CHF 182'240 + CHF 45'000
2013	Lithuanian President Antanas Smetona Scholarship , one year, for excellent grades	

// TEACHING

2026	Workshop, HSLU, Lucerne, Switzerland. compas_wood , advances in timber construction
2025	AA Visiting School Seoul, South Korea. COMPAS Python framework courses
2025	Workshop, Zaha Hadid Architects, London, UK. C++ / Python bindings for developers
2025	Workshop, Calgary, Canada. COMPAS basics, RhinoVAULT form finding, fabrication
2025	Support team, AEC Hackathon, Zurich, Switzerland. COMPAS installation and new features
2024	Workshops, IBOIS, EPFL, IASS 2024, Lausanne, Switzerland. compas_wood , BIM with COMPAS.
2023–2025	Course assistant, BRG, ETHZ, Zurich, Switzerland. Computational Structural Design (CSD I), graphic statics with COMPAS AGS
2020–2022	Co-taught course, EPFL, Lausanne, Switzerland. Intro to Computational Architecture (AR-327)
2020–2021	Master's thesis supervision, EPFL, Lausanne, Switzerland. Maxim Andrist: theory, project, robotic cutting, thesis record
2020–2021	ENAC Week, IBOIS + ALICE, EPFL, Lausanne, Switzerland. “Land of Thousand Dance”, reused timber.
2017–2022	Studio teaching, EPFL, Lausanne, Switzerland. MA1, MA2 and BA/MA studios, project reviews
2017–2022	CNC & robot training, IBOIS, EPFL, Lausanne, Switzerland. Raccoon, Maka 5-axis CNC, ABB robots
2017, 2019	Workshops, VAA, Vilnius, Lithuania. Graph-based aggregations, 1:1 corrugated cardboard shell
ONLINE	Teaching resource, EPFL, Lausanne, Switzerland. Advanced Timber Plate Structural Design (timber joinery generation)

// PUBLICATIONS

2025	Vestartas P. , Füssler L., Rad A. R., Lee D. S., van Mele T., Block P. <i>Exploring the Potential of Funicular Timber Floors</i> . ICSA, Antwerp, Belgium.
2022	Settimi A., Vestartas P. , Gamero J., Weinand Y. <i>Cockroach: an open-source tool for point cloud processing in CAD</i> . CAADRIA, Sydney, Australia. Best Paper Award – Runner-Up. Rogean N., Rezaei Rad A., Vestartas P. , Latteur P., Weinand Y. <i>A Collaborative Workflow to Automate the Design, Analysis, and Construction of Integrally-Attached Timber Plate Structures</i> . CAADRIA, Sydney, Australia.
2021	Vestartas P. , Rezaei Rad A., Weinand Y. <i>Robotically-Fabricated Nexorades from Whole Timber</i> . fib Symposium on the Conceptual Design of Structures, Switzerland. Rezaei Rad A., Burton H., Rogean N., Vestartas P. , Weinand Y. <i>A framework to automate the design of digitally-fabricated timber plate structures</i> . Computers & Structures.
2020	Vestartas P. , Weinand Y. <i>Joinery Solver for Whole Timber Structures</i> . WCTE 2021, Santiago, Chile. Vestartas P. , Weinand Y. <i>Laser Scanning with Industrial Robot Arm for raw wood Fabrication</i> . ISARC 2020, Kitakyushu, Japan, pp. 773–780. Vestarte L., Vestartas P. , Kucinkas R. <i>Corrugated Cardboard Shell: A Pavilion Project of An Architectural Workshop</i> . Advances in Architectural Geometry (AAG).
2019	Vestartas P. , Rogean N., Gamero J., Weinand Y. <i>Modelling Workflow for Segmented Timber Shells using Wood-wood Connections</i> . Design Modelling Symposium, Berlin, Germany, pp. 596–607. Vestartas P. , Palletier L., Nakad M. T., Rad A. R., Weinand Y. <i>Segmented Spiral Using Inter-Connected Timber Elements</i> . IASS Symposium, Barcelona, Spain. Nguyen A. C., Himmer B., Vestartas P. , Weinand Y. <i>Performance Assessment of Double-Layered Timber Plate Shells using Alternative Structural Systems</i> . IASS Annual Symposium, Barcelona, Spain. Nguyen A. C., Vestartas P. , Weinand Y. <i>Design framework for the structural analysis of free-form timber plate structures using wood-wood connections</i> . Automation in Construction.
2017	Ayres P., Vestartas P. , Thomsen M. R. <i>Enlisting Clustering and Graph-Traversal Methods for Cutting Pattern and Net Topology Design in Pneumatic Hybrids</i> . Design Modelling Symposium, Paris, France. Vestartas P. , Heinrich M. K., Zwierzycki M., Leon D. A., Cheheltan A., La Magna R., Ayres P. <i>Design Tools and Workflows for Braided Structures</i> . Design Modelling Symposium, Paris, France. Zwierzycki M., Vestartas P. , Heinrich M. K., Ayres P. <i>High Resolution Representation and Simulation of Braiding Patterns</i> . ACADIA 2017, Cambridge, MA, USA.
BOOKS	2022: <i>Les cahiers de l'Ibois 3 / Ibois Notebooks 3</i> . Introductory text and images, with Berthier S. et al. 2021: <i>Structural design methodology in Integrally-Attached Timber Plate structures</i> . With Rezaei Rad A., in Design of Integrally-Attached Timber Plate Structures, Routledge. 2020: R. Kucinkas (ed.), <i>Digital Architecture at the VAA</i> , 2016: D. Kohler, <i>The Mereological City</i> , 2014: <i>Best graduation projects of architecture students in Baltic states</i>